

5	(a)	region/area where a charge experiences a force	B1	[1]
	(b)	(i) left-hand sphere (+), right-hand sphere (–)	B1	[1]
		(ii) 1 correct region labelled C within 10 mm of central part of plate otherwise within 5 mm of plate	B1	[1]
		2 correct region labelled D area of field not included for (b)(ii)1	B1	[1]
	(c)	(i) arrows through P and N in correct directions	B1	[1]
		(ii) torque = force × <u>perpendicular</u> distance (between forces)	C1	
		$= 1.6 \times 10^{-19} \times 5.0 \times 10^4 \times 2.8 \times 10^{-10} \times \sin 30$		
		$= 1.1 \times 10^{-24} \text{ N m}$	A1	[2]

6 (a) (i) $P = VI$

C1